

Identification of Undocumented Meat Samples Seized at Heathrow Airport

1935263

Abstract

DNA barcoding provides a rapid approach for morphologically compromised sample identification, which is crucial given potential links to illegal trade and public health. This report aimed to resolve the species of four undocumented meat samples from Heathrow, and for this a pipeline was constructed that processed raw sequence data through to phylogenetic tree construction.

Phylogenetic analysis placed the samples (A-D) within four vertebrate clades: Balaeidae (A), Monodontidae (B), *Thunnus* spp. (C), and Testudines (D). Species level resolution was limited due to poor sequencing reads and short barcoding regions, but where possible BLAST and phylogeny together suggested the following candidate species: North Pacific right whale (A), beluga or narwhal (B), tuna species (C), and leatherback sea turtle (D). Samples A and D raised conservation concerns from assessments of the candidate species, with potential illegal trade network involvement, although geographic origin is required to assess conservation concerns fully.

Despite limited species level resolution, this analysis demonstrated DNA barcoding and phylogenetics as useful tools for conservationists and border officials in identifying undocumented meat samples. Improvements to sequence quality and barcode length are suggested to aid species level resolution and enable a comprehensive assessment of conservation status and trade legality.

Introduction

The illegal wildlife trade drives endangered species towards extinction, threatening ecosystems and reinforcing a detrimental feedback loop between species rarity and value [1]. Species identification of undocumented meat samples is key in public health, but can also uncover networks carrying out illegal imports of endangered species, aiding conservation efforts by providing routes for intervention.

If the morphology of a sample is compromised, DNA barcoding - comparing genetic sequences in a sample against known references - can aid in resolving species identity [2, 3]. Mitochondrial genes are used for barcoding because they occur in high copy numbers relative to nuclear DNA and typically lack recombination, enabling species identification [4]. Cytochrome c oxidase I (COI) is the primary barcode marker in Metazoa, and COI sequences are abundant in databases such as BOLD and NCBI, permitting barcoding against a diverse set of organisms [5, 6].

This analysis aims to resolve the species of four meat samples intercepted by UK Border Force officers at Heathrow Airport.

Methods

Pre-Processing

FASTQ sequence files were extracted from `samples.zip` before undergoing quality control. Samples were concatenated and `seqtk` [7] converted to FASTA, masking bases with quality scores < 20 to ensure bases were called at a confidence $\geq 99\%$, reducing BLAST false positive likelihood.

BLAST

BLAST [8] searches were performed against the NCBI Nucleotide database using `blastn` [9]. As samples were labeled without genomic region context, parts were treated as separate fragments and BLAST results were split by part. These were sorted and the top four hits selected. `sampleD_part3` returned *Homo sapiens*, consistent with human contamination, and was removed.

Accessions were downloaded with `efetch` [10] and trimmed to the query region using `seqtk`. FASTA sequences were concatenated by part into three complete files using `seqkit` [11]. *Selachii* were included as the outgroup.

Translation and Alignment

Sequences were translated using Biopython [12] and the vertebrate mitochondrial code. MUSCLE5 [13] generated multiple sequence alignments, and `trimAl` [14] removed poorly aligned regions. BLAST results indicated all parts originated from COI, so the alignments were concatenated into a supermatrix alignment with `catfasta2phym` [15], retaining the three partitions.

Tree Building

Phylogenetic inference was performed using maximum likelihood in IQ-TREE3 [16] with ModelFinder Plus and 1000 ultrafast bootstrap replicates to estimate the tree with the best-fitting substitution model with statistical node support. The resulting tree was rooted to the outgroup and visualised using Python with `ete3` [17].

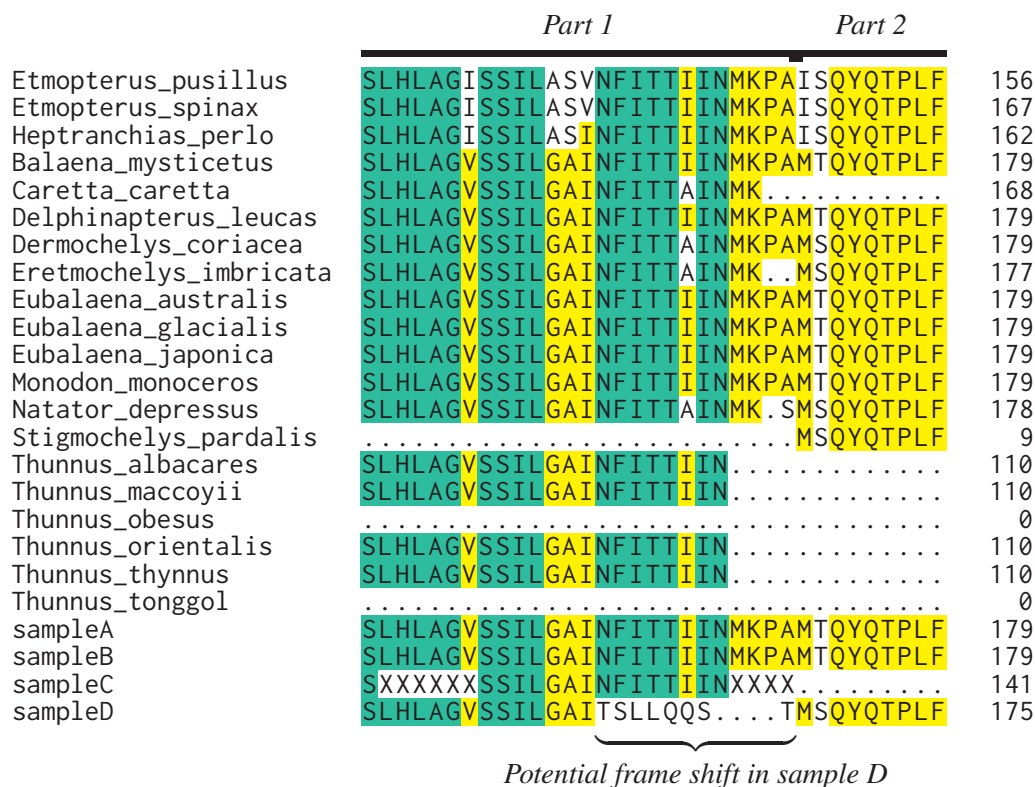
Results

BLAST searches suggested that the samples corresponded to COI and provided preliminary identifications. No hits returned from `sampleC_part2`, reducing usable sequence length (Table 1).

The phylogenetic tree (Fig. 2) recovered four resolved vertebrate clades: Cetaceans, Testudines, Scombridae, and outgroup *Selachii*.

Table 1: Top BLAST hit across all samples

Sample	Part	Species	% Identity	e-value
A	1	<i>Eubalaena japonica</i>	97.5	0.0
A	2	<i>Eubalaena japonica</i>	99.6	0.0
A	3	<i>Eubalaena japonica</i>	93.4	7.0e-160
B	1	<i>Delphinapterus leucas</i> ; <i>Monodon monoceros</i> (tie)	92.3	0.0
B	2	<i>Delphinapterus leucas</i>	92.8	0.0
B	3	<i>Monodon monoceros</i>	89.2	0.0
C	1	<i>Thunnus thynnus</i> ; <i>Thunnus orientalis</i> (tie)	81.7	3.7e-103
C	2	—	—	—
C	3	<i>Thunnus thynnus</i>	89.2	1.2e-157
D	1	<i>Dermochelys coriacea</i>	98.3	0.0
D	2	<i>Dermochelys coriacea</i>	96.2	0.0
D	3	<i>Homo Sapiens</i>	100	6.3e-46



[] non-conserved
 [X] ≥ 50% conserved
 [X] ≥ 85% conserved

Figure 1: Window of supermatrix alignment

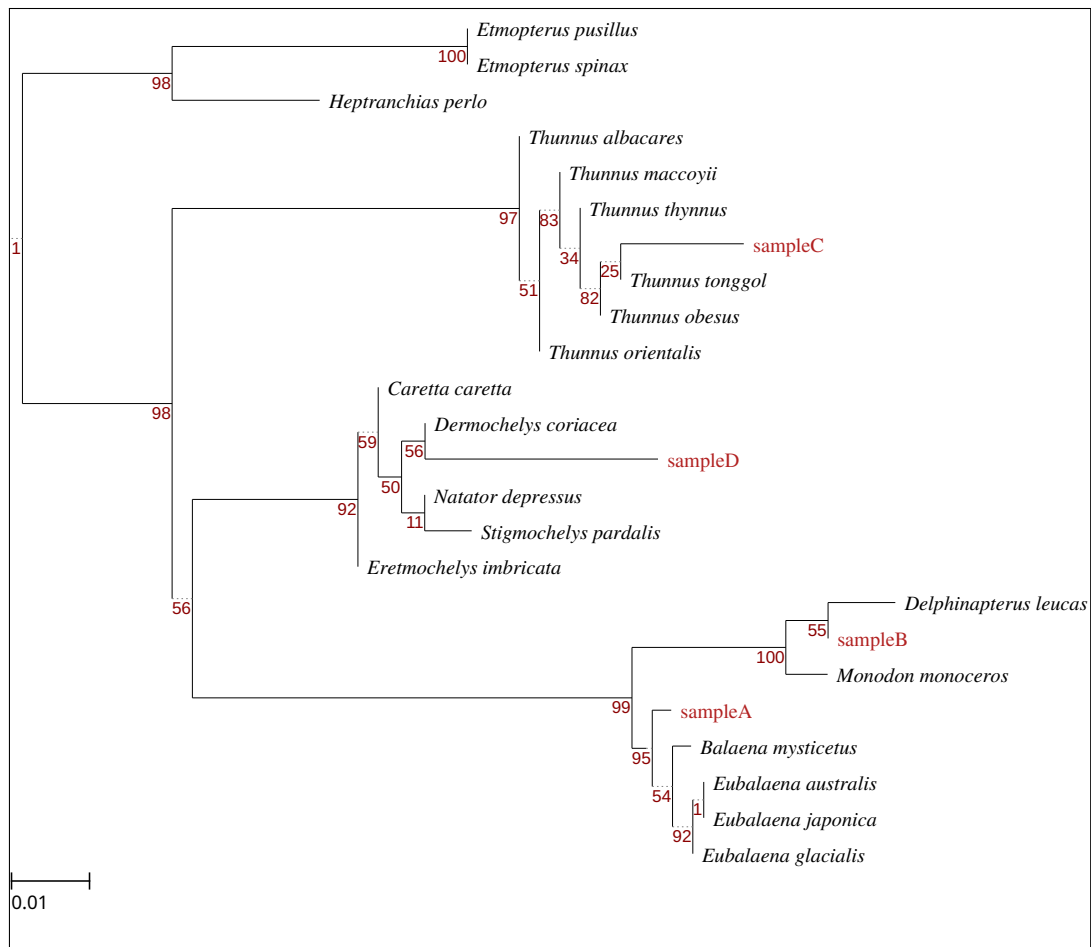


Figure 2: Phylogenetic tree with placement of unknown meat samples and closest BLAST hits. Scale bar = 0.01 substitutions per site. Bootstrap support shown at nodes.

Sample A placement was strongly supported within Balaenidae (95% bootstrap), but species level resolution was poor. BLAST consistently returned *Eubalaena japonica* as the top hit with a high percentage identity across all parts, suggesting North Pacific right whale as the leading candidate.

Sample B placement within Monodontidae was fully supported (100%), but internal support was weak (55%). Only *Delphinapterus leucas* and *Monodon monoceros* exist in this family, therefore sample B represents one of these species.

Sample C placement was strongly supported within *Thunnus* (97%), with low-moderate internal node support. BLAST suggested *Thunnus thynnus*, but this is not confirmed by the phylogeny. Sample C may only be assigned to the *Thunnus* genus.

Sample D placement within Testudines was supported (92%), with low support within (11-59%). Sample D clustered with *Dermochelys coriacea*, consistent with BLAST, suggesting Leatherback sea turtle as the leading candidate.

Discussion

Order level placement was strongly supported with confident assignment to major vertebrate groups. However, species confidence was low and relied on BLAST compliance with phylogeny for candidate species proposals.

The COI fragment for barcoding is only $\approx 650\text{bp}$ which makes species inference sensitive to sequencing errors, with low quality reads providing limited phylogenetic signal [3]. Low quality and ambiguous calls appeared within all samples, reducing usable bases and likely weakening discriminatory power and reducing bootstrap support. Exploratory alignment (Fig. 3) indicated a potential sequencing misread leading to the apparent frame shift in Fig. 1. Adding one ambiguous base increased *Dermochelys coriacea*/sample D clade support from 56% to 72% (Figs. 4, 5), illustrating the impact of data quality on phylogenetic confidence. Furthermore, reference sequences from GenBank do not require strict validation and must be interpreted cautiously for species identification [4, 18].

Sample A candidate, *Eubalaena japonica*, is listed as ‘Endangered’ [19]. Presence of this sample raises serious conservation concerns if confirmed, and could reflect trade of a vulnerable whale species. Both sample B candidates are listed under ‘Least Concern’ [20, 21]. Their presence may indicate by-catch from fisheries in or near conservation sites, but without sample origin these implications cannot be assessed fully. *Thunnus* assessments range from ‘Least Concern’ (*Thunnus albacares*) to ‘Endangered’ (*Thunnus maccoyii*) [22, 23]. Without species level identification the conservation implications of sample C cannot be assessed. Sample D candidate, *Dermochelys coriacea*, is listed as ‘Vulnerable’ [24]. This species inhabits protected areas across its range, as such the presence of this sample raises conservation concerns and may reflect illegal trade, depending on the origin of the sample.

Overall, COI barcoding here resolved order and family level identity but lacked species level resolution. Longer barcode regions and multiple reads of sample sequences could reduce misreads while increasing resolution among recently diverged species. The use of curated reference databases, such as BOLD, would increase confidence in reference sequences [18]. These suggestions will allow more confident conservation claims for the meat samples. Nevertheless, the detection of candidate species from undocumented meat illustrates the value of DNA barcoding in the context of the illegal wildlife trade.

References

- [1] Liam J Hughes, Oscar Morton, Brett R Scheffers, and David P Edwards. The ecological drivers and consequences of wildlife trade. *Biological Reviews*, 98(3):775–791, 2023.
- [2] Pilar A Haye, Nicolás I Segovia, Raúl Vera, María de los Ángeles Gallardo, and Cristian Gallardo-Escárate. Authentication of commercialized crab-meat in chile using dna barcoding. *Food Control*, 25(1):239–244, 2012.
- [3] Sarfraz Ahmed, Muhammad Ibrahim, Chanin Nantasenamat, Muhammad Farrukh Nisar, Aijaz Ahmad Malik, Rashem Waheed, Muhammad Z Ahmed, Suvash Chandra Ojha, and Mohammad Khurshed Alam. Pragmatic applications and universality of dna barcoding for substantial organisms at species level: a review to explore a way forward. *BioMed Research International*, 2022(1), 2022.

- [4] Nick Dawnay, Rob Ogden, Ross McEwing, Gary R Carvalho, and Roger S Thorpe. Validation of the barcoding gene *coi* for use in forensic genetic species identification. *Forensic science international*, 173(1):1–6, 2007.
- [5] Teresita M Porter and Mehrdad Hajibabaei. Over 2.5 million *coi* sequences in genbank and growing. *PloS one*, 13(9):e0200177, 2018.
- [6] Sujeevan Ratnasingham, Catherine Wei, Dean Chan, Jireh Agda, Josh Agda, Liliana Ballesteros-Mejia, Hamza Ait Boutou, Zak Mohammad El Bastami, Eddie Ma, Ramya Manjunath, et al. Bold v4: A centralized bioinformatics platform for dna-based biodiversity data. In *DNA barcoding: Methods and protocols*, pages 403–441. Springer, 2024.
- [7] Heng Li. seqtk: Toolkit for processing sequences in FASTA/Q formats . <https://github.com/lh3/seqtk>, 2012.
- [8] Stephen F Altschul, Warren Gish, Webb Miller, Eugene W Myers, and David J Lipman. Basic local alignment search tool. *Journal of molecular biology*, 215(3):403–410, 1990.
- [9] Christiam Camacho, George Coulouris, Vahram Avagyan, Ning Ma, Jason Papadopoulos, Kevin Bealer, and Thomas L Madden. Blast+: architecture and applications. *BMC bioinformatics*, 10(1):421, 2009.
- [10] Jonathan Kans. Entrez direct: E-utilities on the unix command line. In *Entrez programming utilities help [Internet]*. National Center for Biotechnology Information (US), 2024.
- [11] Wei Shen, Shuai Le, Yan Li, and Fuquan Hu. Seqkit: a cross-platform and ultrafast toolkit for fasta/q file manipulation. *PloS one*, 11(10):e0163962, 2016.
- [12] Peter JA Cock, Tiago Antao, Jeffrey T Chang, Brad A Chapman, Cymon J Cox, Andrew Dalke, Iddo Friedberg, Thomas Hamelryck, Frank Kauff, Bartek Wilczynski, et al. Biopython: freely available python tools for computational molecular biology and bioinformatics. *Bioinformatics*, 25(11):1422, 2009.
- [13] Robert C Edgar. Muscle5: High-accuracy alignment ensembles enable unbiased assessments of sequence homology and phylogeny. *Nature communications*, 13(1):6968, 2022.
- [14] Salvador Capella-Gutiérrez, José M Silla-Martínez, and Toni Gabaldón. trimal: a tool for automated alignment trimming in large-scale phylogenetic analyses. *Bioinformatics*, 25(15):1972–1973, 2009.
- [15] Johan Nylander. catfasta2phyml – Concatenate FASTA alignments to PHYML, PHYLIP, or FASTA format. <https://github.com/nylander/catfasta2phyml>, 2010.
- [16] Thomas KF Wong, Nhan Ly-Trong, Huaiyan Ren, Hector Baños, Andrew J Roger, Edward Susko, Chris Bielow, Nicola De Maio, Nick Goldman, Matthew W Hahn, et al. Iq-tree 3: Phylogenomic inference software using complex evolutionary models. 2025.
- [17] Jaime Huerta-Cepas, François Serra, and Peer Bork. Ete 3: reconstruction, analysis, and visualization of phylogenomic data. *Molecular biology and evolution*, 33(6):1635–1638, 2016.
- [18] Kelly A Meiklejohn, Natalie Damaso, and James M Robertson. Assessment of bold and genbank—their accuracy and reliability for the identification of biological materials. *PloS one*, 14(6):e0217084, 2019.
- [19] J. G. Cooke and P. J. Clapham. *Eubalaena japonica*. <https://dx.doi.org/10.2305/IUCN.UK.2018-1.RLTS.T41711A50380694.en>, 2018. Accessed 07 December 2025.

- [20] L. Lowry, K. Laidre, and R Reeves. Monodon monoceros. <https://dx.doi.org/10.2305/IUCN.UK.2017-3.RLTS.T13704A50367651.en>, 2017. Accessed 07 December 2025.
- [21] L. Lowry, K. Laidre, and R Reeves. Delphinapterus leucas. <https://dx.doi.org/10.2305/IUCN.UK.2017-3.RLTS.T6335A50352346.en>, 2017. Accessed 07 December 2025.
- [22] B.B. Collette, A. Boustany, W. Fox, J. Graves, M. Juan Jorda, and V Restrepo. Thunnus albacares. <https://dx.doi.org/10.2305/IUCN.UK.2021-2.RLTS.T21857A46624561.en>, 2021. Accessed 07 December 2025.
- [23] B.B. Collette, A. Boustany, W. Fox, J. Graves, M. Juan Jorda, and V Restrepo. Thunnus maccoyii. <https://dx.doi.org/10.2305/IUCN.UK.2021-2.RLTS.T21858A170082633.en>, 2021. Accessed 07 December 2025.
- [24] B.P. Wallace, M. Tiwari, and M Girondot. Dermochelys coriacea. <https://dx.doi.org/10.2305/IUCN.UK.2013-2.RLTS.T6494A43526147.en>, 2013. Accessed 07 December 2025.
- [25] Kazutaka Katoh and Daron M Standley. Mafft multiple sequence alignment software version 7: improvements in performance and usability. *Molecular biology and evolution*, 30(4):772–780, 2013.

Supplementary Data

Supplementary data and scripts are found in the following GitHub repository: https://github.com/archiebenn/BIOLM0051_analysis.git

Appendix: Exploratory Figures

Dermochelys_c	441	CACCTAGCTGGTGTTCATCAATTTTAGGAGCTATTA	490
sampleD_part1	451	CACCTAGCTGGTGTTCATCAATTTTAGGAGCTATT-AC	499



Figure 3: MAFFT [25] Needleman-Wunsch alignment. Gap indicates a possible misread during sequencing.

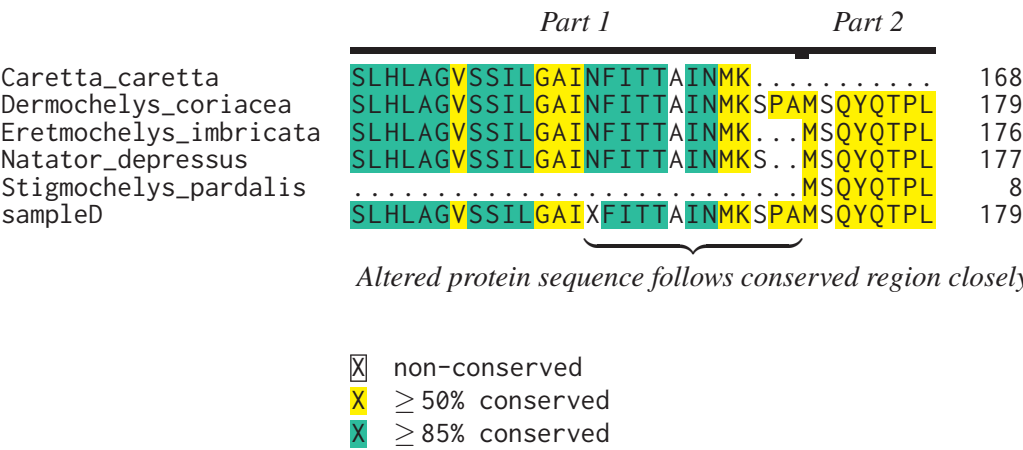


Figure 4: Alignment after inserting one ambiguous base (N) in sampleD_part1

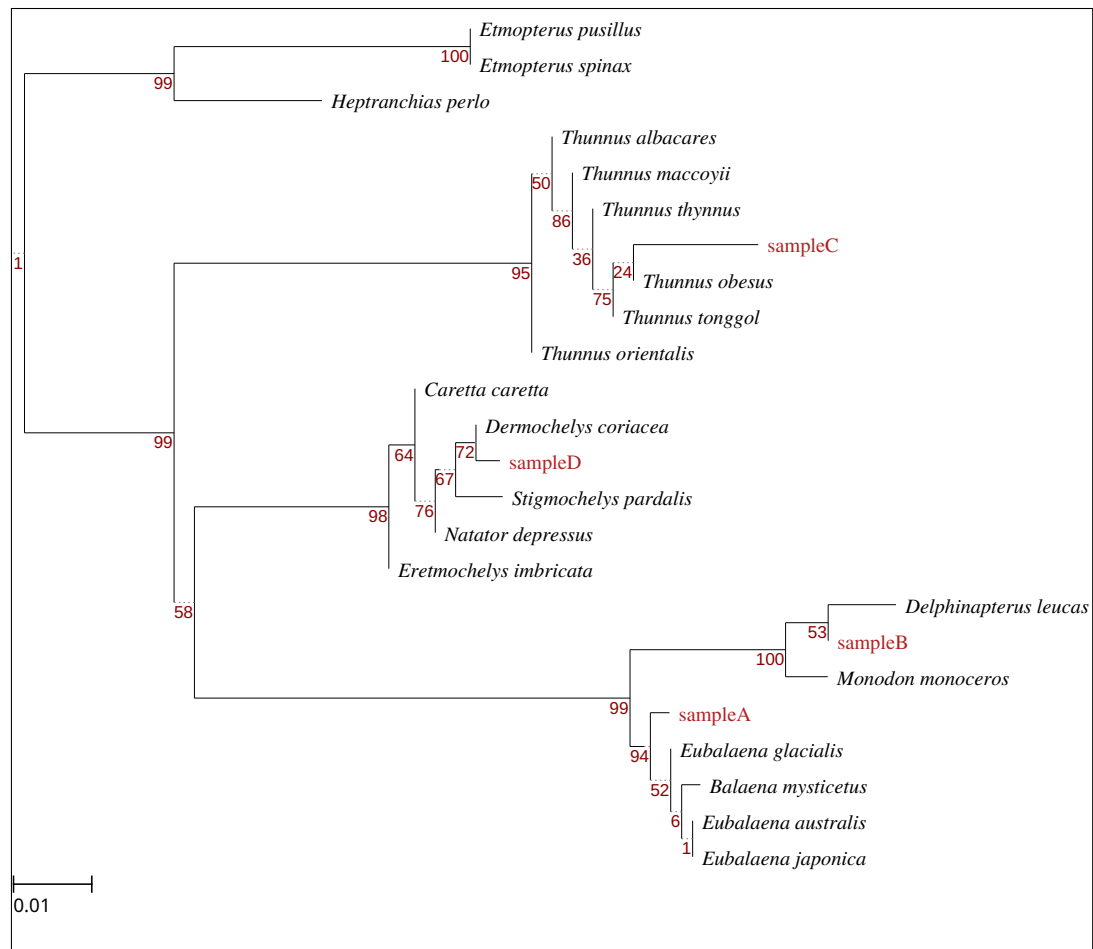


Figure 5: Phylogenetic tree resulting from altered alignment (Fig. 4).